Enriching the Search Experience with Analytics

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Overview

R&D

Goals

- Intelligent integration of structured data (e.g. data in relational databases), unstructured data (e.g. human-produced reports), and semi-structured data (e.g. Excel spreadsheets, RSS feeds, etc.)
- Use off-the-shelf analytical techniques to improve information retrieval and exploration process for users

Unstructured Data

- Using Solr, indexing unstructured data is easy
- Use Tika to extract text content
- Store extracted text in Solr index along with additional document metadata
- Provided faceted search experience over the data

Enriching the search experience with analytics

Ability to index unstructured data is straight forward

Use tika to extract text content. Basically just index the words

Store extracted text in solr index along with additional metadata such as document type, data, user, …

Provide faceted search experience over the data

Major challenges are volume and size of data

Also dealing with the types of text and providing appropriate analyzers and such

What if we could do more than pull out the words?

Extract geospatial, temporal, person, organization entities

Pull out topics, find relationships between documents

Provide sentiment analysis

Store the results directly in the solr index!

Semantically enrich the solr index with this information

Now we can provide a richer experience

Find documents relevant to a geospatial or temporal constraint

Allows us to have more "information" on the documents

How are we doing this?

Pipeline processing of documents

Tika text extraction

Named entity extraction

Use geo gazetteer to help with geo extraction

Use rules for location matching (lat/lon, utm, …)

Rules for temporal extraction

Mine structured data to find things like organizations, persons, and other entities

Geohash to provide efficient query of geo information

Clustering algorithms to find "similar" content

User interfaces

Search and facetted navigation

What if we could do more than pull out the words?

SAN JOSE, Calif.—A federal judge overseeing the high-stakes patent trial between Apple Inc. and Samsung Electronics Co. requested Wednesday that the chief executives of both companies talk one more time to try to settle the case before it is handed to a jury.

Speaking before jurors entered the courtroom for another day of proceedings, Judge Lucy Koh said she "sees risks here for both sides" in handing the case—which centers on the companies' smartphones—to the nine-member jury.

Lawyers for both companies agreed that the chiefs would attempt to talk. Judge Koh requested, and the companies agreed to, a similar in-person meeting between the CEOs before the trial began last month. No settlement was reached.

South Korea's Samsung, which Apple sued in 2011, on Wednesday was set to continue presenting its case, seeking to refute Apple's charges of patent infringement and prove its own infringement charges against the Silicon Valley company.

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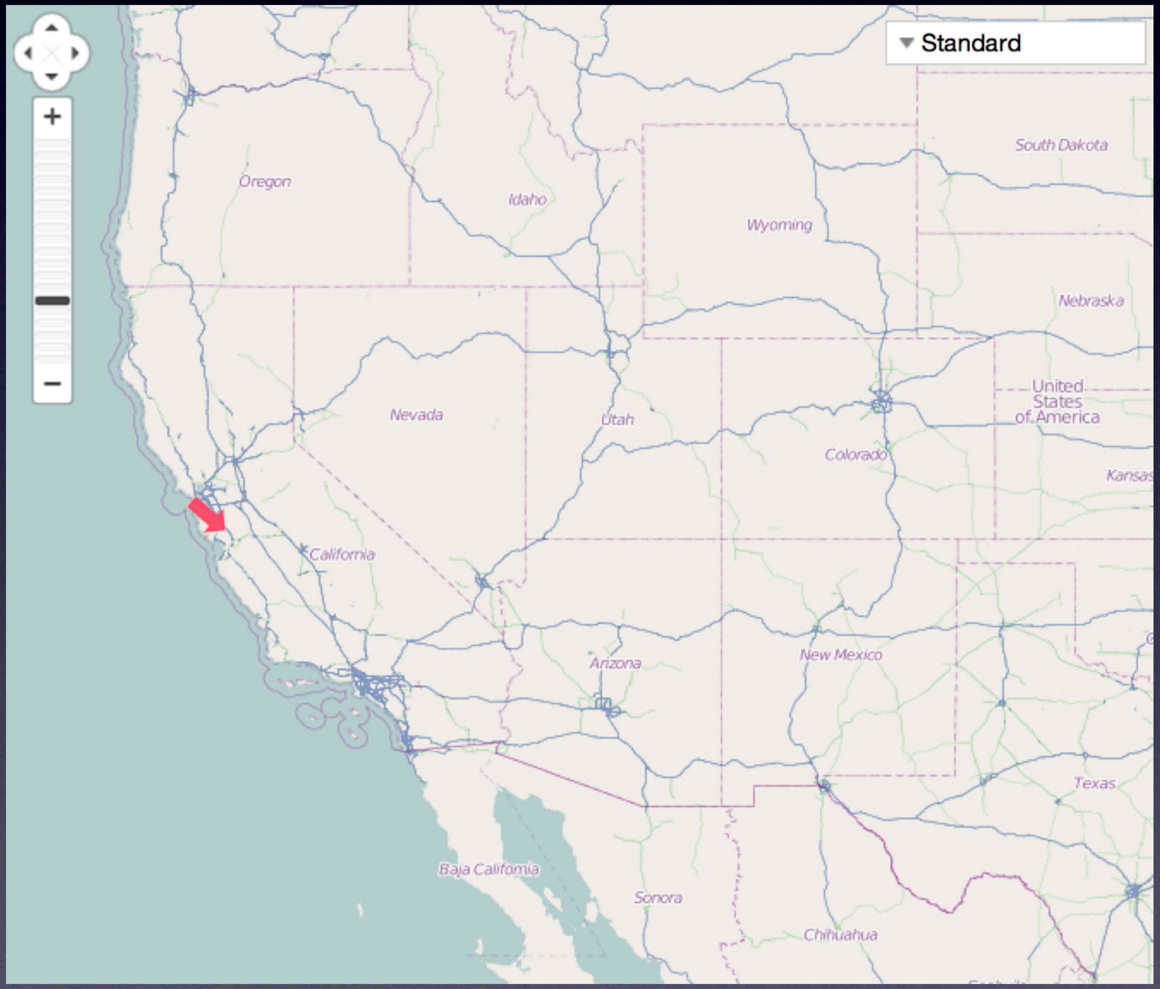
Extraction steps

1. Extract "entities" from document using named-entity recognition, aka entity extraction - geospatial (places), temporal (dates), people, organizations, proper names, etc.
2. Pull out topics, find relationships between documents
3. Store the results directly in Solr index!
4. Semantically enrich the Solr index with this information

Richer Experience

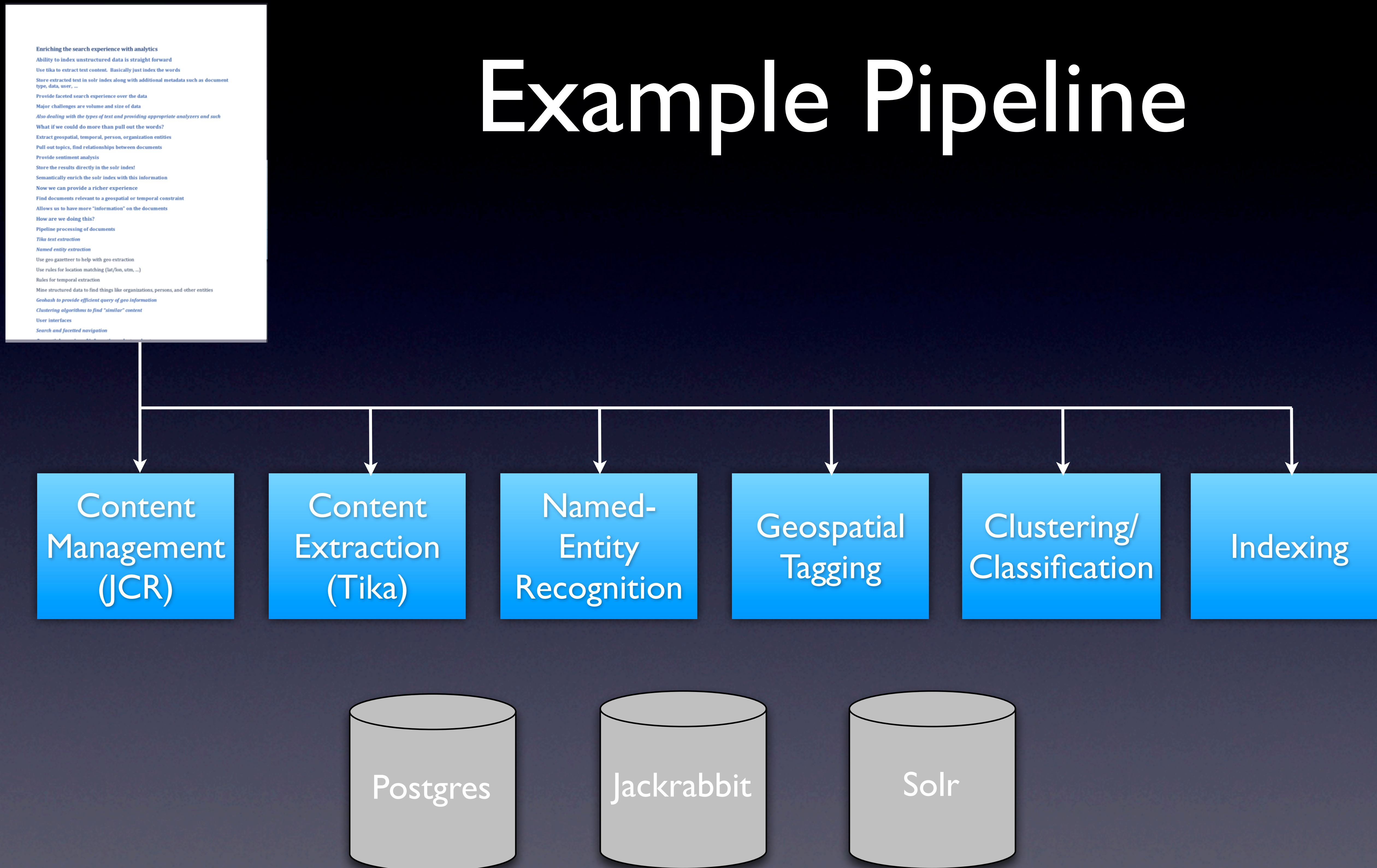
- Example: find documents relevant to a provided temporal or geospatial constraint
- Allows us to have more “information” on the documents



How is this accomplished?

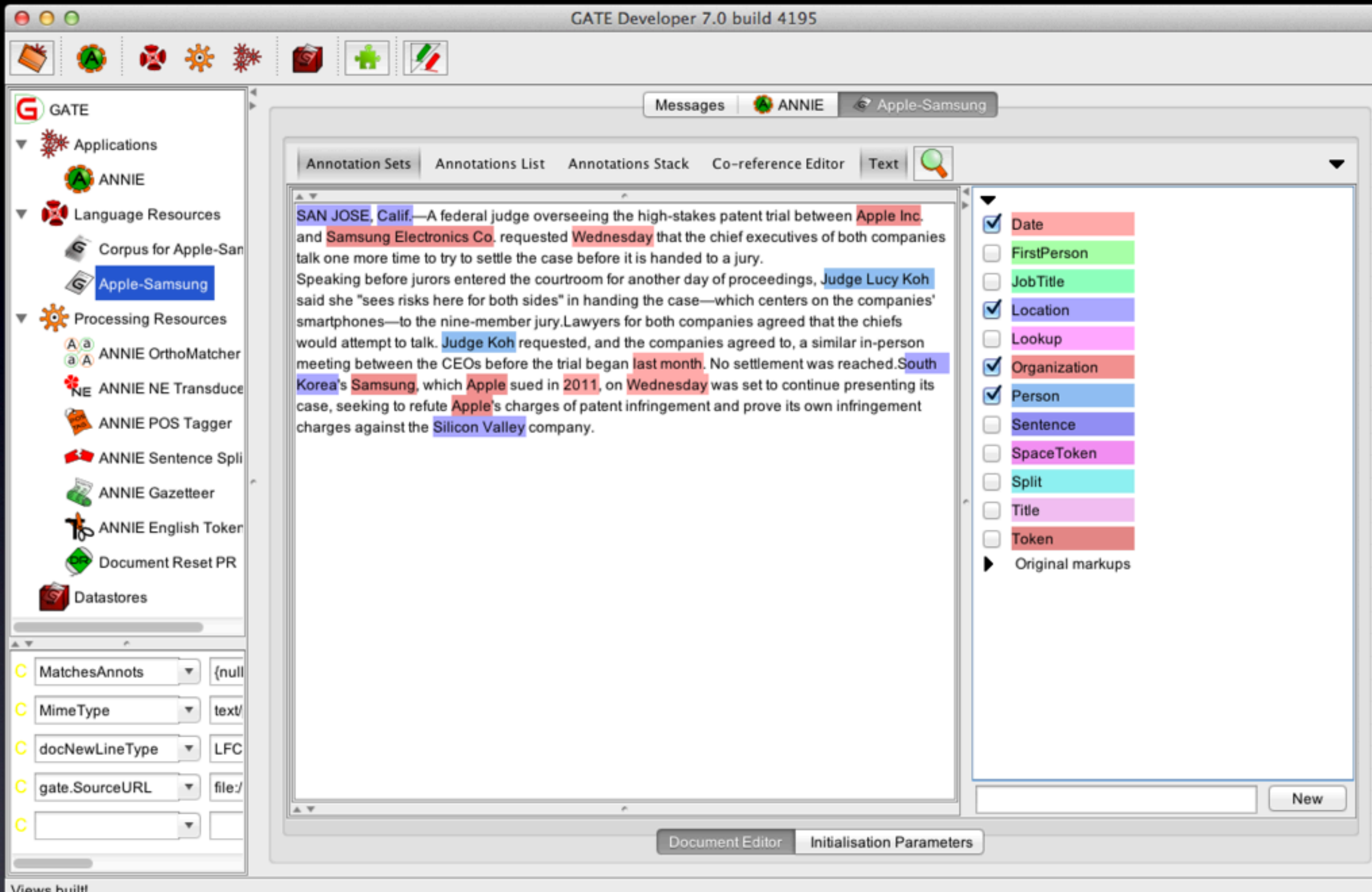

562

Example Pipeline



Named-Entity Recognition

- Use standard text processing techniques to identify tokens that are likely proper nouns
- Use regular expressions for matching coordinates (lat/lon, UTM, MGRS, etc.)
- Use GeoNames lookup to validate extracted geospatial entities


GATE Developer 7.0 build 4195
Messages
ANNIE
Apple-Samsung
GATE
Applications
ANNIE
Language Resources
Corpus for Apple-San
Apple-Samsung
Processing Resources
ANNIE OrthoMatcher
ANNIE NE Transduce
ANNIE POS Tagger
ANNIE Sentence Spli
ANNIE Gazetteer
ANNIE English Toker
Document Reset PR
Datastores
MatchesAnnots
{null
MimeType
text/
docNewLineType
LFC
gate.SourceURL
file:/
Annotation Sets
Annotations List
Annotations Stack
Co-reference Editor
Text
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Date
FirstPerson
JobTitle
Location
Lookup
Organization
Person
Sentence
SpaceToken
Split
Title
Token
Original markups
New
Document Editor
Initialisation Parameters
Views built!

User Interface

- Search and faceted navigation
- Geospatial overview of information - heatmaps, clusters, etc.
- Temporal overview of information - timelines, histograms, time-series analysis
- How to combine structured data with unstructured data?

Where are we at?

- Currently using Solr 3.6, transitioning to Solr 4.0 beta (stick around for next presentation)
- Experimenting with Solr geospatial capabilities
- Trying different named-entity recognition approaches
  - GATE/ANNIE - rules based
  - Statistical and machine learning based approaches that incorporate user feedback

Summary

- Trying to develop standard implementation so that all of the standard “good stuff” that Solr can do, we *actually do*
- Incorporate off-the-shelf analytical techniques (information extraction, machine learning, etc.) to pick off “low-hanging fruit” of enriching the search experience
- Allow us to focus on the truly hard problems - is this the same person, is this the correct date/location, etc.?

Questions?

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